



Role and value of urban forests in carbon sequestration: review and assessment in Indian context

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Abstract

The role of urban forests in carbon capture and sequestration has been reviewed in a global context. An interpretive analysis was carried out using normalized difference vegetation index (NDVI) and allometric relationships in 5 cities of India, viz. Navi Mumbai, Nagpur, Bengaluru, Leh and Delhi, to evaluate the economic importance of urban green spaces in sequestering carbon emissions and mitigating severe climatic impacts. NDVI maps were prepared using data from Landsat 7 and Landsat 8 satellite sensors. Bengaluru showed highest carbon sequestration (141.83 MT), and carbon sequestration in the city of Leh was comparatively very low (1.51 MT) as it is a cold desert with a very low urban forest cover. The valuation of urban green spaces in the selected cities was done by using market price method (MPM), value transfer method (VTM), and social cost of carbon method (SCC). The carbon sequestration valuation ranged from 55 to 5164 million\$ for estimation using MPM and from 69.69 to 6537.15 million \$ for SCC, whereas for VTM it ranged from 1575.06 to 147,736.94 million INR. The SCC approach used for valuation showed the highest cost of CO₂ sequestered, while the MPM provided with the lowest value for all the five cities. Highest carbon sequestration and valuation values were observed in Bengaluru followed by Delhi. This study also points to the need of developing region-specific allometric relationships for better accuracy in quantifying carbon sequestration potential and associated economic value of urban forests.

Keywords NDVI · Terrestrial carbon sequestration · Urban green space · Value transfer method · Social cost of carbon

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1 Introduction

The possibility for long-term changes in climate and temperature patterns due to the imbalance in the carbon cycle has become one of the major environmental concerns for our planet (Solomon et al., 2009). Carbon contributes to the global cycle through reactions inside and interchange among the primary global reservoirs, which are referred to as carbon pools (Fig. 1). The carbon cycle is significant because carbon-containing substances in the atmosphere influence the earth's climate. Carbon dioxide (CO₂) levels in the atmosphere have been responsible for more than half of the recent climate warming (IPCC, 2014). Carbon dioxide concentrations in the atmosphere have climbed from around 278 parts per million (ppm) in 1750 to 410 ppm now since the start of the industrial era (Forster et al., 2007; Tans & Keeling, 2012).

The scientific community have been dealing with the problem of increased CO₂ in the atmosphere on two fronts: (1) reducing the carbon emissions so as to limit the corresponding global temperature increase below a certain level and (2) reducing the accumulation of CO₂ in the atmosphere by carbon capture and sequestration (CCS). Major challenges in reducing CO₂ emissions are not technical, but rather political which demand intergovernmental cooperation among culturally different and economically competing nations. CCS, from the other hand, refers to both natural and manmade processes that remove CO₂ from the atmosphere or divert it from the sources of emissions and store it in geologic formations, oceans, and terrestrial environments. Among various CCS strategies (Nogia et al., 2016), terrestrial carbon sequestration has been widely accepted because of its advantage of long-term storage and sustainability.

Terrestrial sequestration is a natural process that involves the storage of CO₂ in soil and vegetation as above- and below-ground biomass (Post et al., 2009). The potential of global forest systems as a sink for carbon has been well studied and documented. From 1990 to 2007, the global forest ecosystem was discovered to have a net carbon sink of 1.11 Gt of carbon per year (Pan et al., 2011). Trees, whether part of a forest system or not, plays an important role in the terrestrial carbon sequestration. Trees that are not part of forest system are categorized as Trees Outside Forests (TOF) and include agroforests, urban and

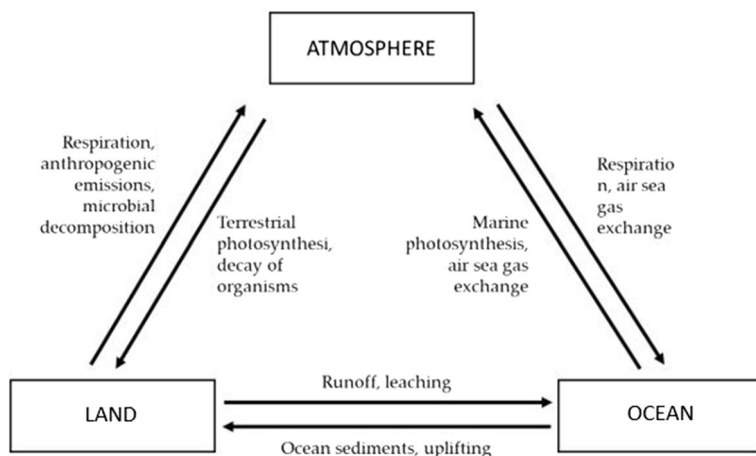


Fig. 1 Exchange of carbon between land, ocean, and atmosphere

rural forests. This paper focuses primarily on the role that urban forests could play in mitigating carbon emissions, in addition to the services they offer such as filtering air pollutants, increasing urban biodiversity, improving social health, regulating water flow, cooling air, etc. As the scope of increasing the area of forest systems across the globe is getting limited due to the increased pressure of urbanization, nurturing urban forests would provide a promising alternate solution to enhance terrestrial carbon sequestration.

2 Urban forests—classification and assessment

Urban forests are defined as individual or group of trees that grow within an urban or sub-urban area. Urban forests can be termed as green infrastructure of a city, which contribute to physiological, sociological, and economic well-being of urban societies. Sustainability of any city depends on optimized interaction among its grey infrastructure (e.g. buildings, roads, utilities, etc.), blue infrastructure (e.g. lakes, ponds, river, etc.), and green infrastructure (e.g. urban forests). Table 1 lists the potential benefits of urban forests in addressing major urban issues.

The urban forests, in general, can be categorized into various types based on the following aspects.

(a) Cause of existence

Depending on how vegetation happened to exist in an urban area, urban forests could be classified into two types, namely

1. *Natural*—remnants of natural forest that existed before urbanization and reduced to patches of trees within and around an urban area.
2. *Planned*—green spaces that have been designed as part of urban planning which later became a full-grown urban forest.
3. *Spatial planning*

Depending on the spatial arrangement within a city, urban forests are classified as,

1. *Sub-urban forests*—Forest that surround an urban area and provide goods and services such as forest products and recreation.

Table 1 Urban forests and their potential benefits

Urban Issue	Potential benefit
Air and noise pollution	It filters the air pollutants and acts as a barrier to reduce noise
Water scarcity	Improves infiltration, reduces storm water runoff
Heat island	provides shade, cools the environment through evapotranspiration, indirectly reduces energy consumption and associated emissions.
Health of society	Provides recreational spaces and improves physical and mental health
Carbon emission	Sequesters carbon, reduces climate impact of urban emissions
Landscape degradation	Prevents soil erosion
Lack of biodiversity	Increases biodiversity

2. *City parks*—large land cover with an area greater than 0.5 ha with facilities for leisure.
3. *Gardens*—small green spaces with an area less than 0.5 ha with facilities for recreation.
4. *Street vegetation*—small tree populations on the sides of the streets, parking lots, squares, etc.
5. *Other green spaces with trees*—agricultural land, open fields, cemeteries, etc.
6. Species diversity

Depending on the type of species found in urban forests, they are classified as:

1. *Native*—forest contain tree species that are naturally originated in the area under consideration.
2. *Exotic*—forest contain tree species which were introduced through human activity, deliberately or by accident.

Local plants have a higher potential for use in carbon sequestration projects than exotic trees because they are better adapted to local conditions, are less damaging to soils, supplement local and indigenous sources of knowledge and culture, and have a greater diversity of species on a landscape. A prior study proved the capacity of mixed-species native plantations to flourish at rates competitive with alien plantings (Montagnini et al., 1998).

Trees in metropolitan areas are often appraised at the city or town level by municipalities. Many cities across the world have their own assessment and monitoring Programme for urban forestry. Four countries (the UK, India, Sweden, and the USA) have done regional or nationwide integrated assessments of urban trees (de Foresta et al. 2013). In few studies, it is found that the low carbon areas (green spaces) are having higher thermal comfort (Bherwani et al., 2021a; Mirza et al., 2022).

In the USA, "Forest on the Edge" is a long-term US Forest Service Programme focused to urban forestry. The initiative published a report that included the major findings of a nationwide assessment of urban forests, which were defined as all publicly and privately held trees inside an urban region, including trees along streets and in backyards, as well as remnant forest stands (Nowak et al., 2010). The study, which employed high-resolution remote sensing data combined with maps of urban areas, did not contain any field measurements and instead focused solely on tree canopy cover and tree density in each county's urban districts.

The Trees in Town II initiative (2004–2008) in the UK involved local, regional, and national organizations and institutions from many sectors, coordinated by the Department for Communities and Local Government. Its goal was to provide current information on England's urban tree supply and urban tree management. The evaluation was based on a preliminary stratification (region, town size, and land-use type), followed by a random selection of field samples spread across all strata. It includes a survey questionnaire that was distributed to all municipal governments in responsibility of city trees. In addition, a combination of high-resolution, remote sensing-derived data and measures of geographical, biophysical, managerial, and socioeconomic aspects in each field sample ensured high-quality qualitative and quantitative results. The "Trees in Town" experiment was partially replicated in Sweden, where an urban forestry assessment was done in 2006 using survey questionnaires distributed to municipal governments (FAO, 2010).

The "TOF urban" evaluation is part of the periodic national forest inventory in India. It collects various geographical, biophysical, and socioeconomic characteristics from randomly selected "urban block" samples. The number of field samples in a city grows in

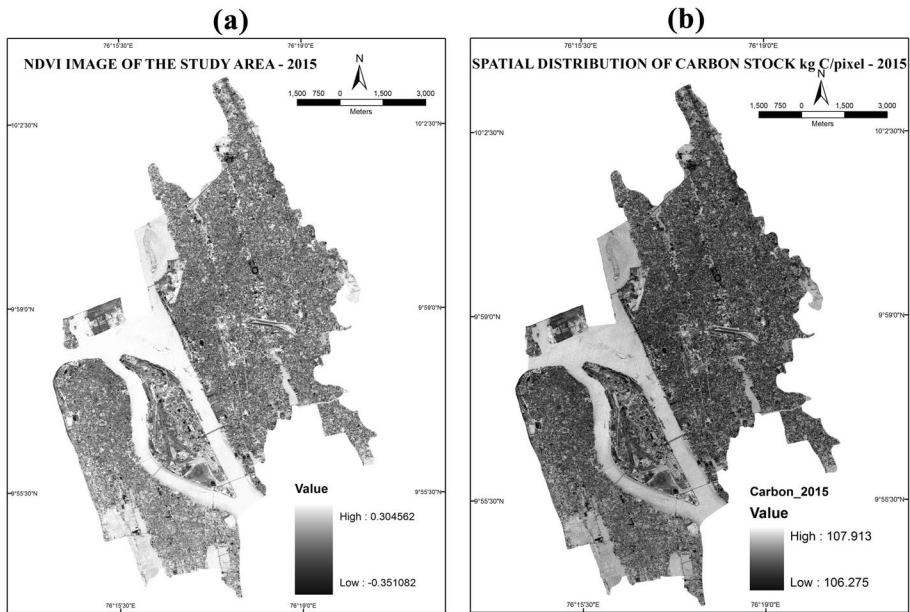


Fig. 2 Carbon storage potential of urban forest in Kochi city, India. **a** NDVI estimated from IRS P6 LISS IV sensor and **b** spatial distribution of carbon stock in the urban area

proportion to the number of blocks. The Indian "TOF urban" inventory, like the "TOF rural" inventory, resulting in a solid and accurate knowledge base on TOF in urban areas on a national scale (FSI, 2002).

3 Carbon storage in urban forests—case studies

Though most of our knowledge of urban trees' carbon storage and sequestration comes from studies conducted in the USA (McPherson, 1994; Nowak & Crane, 2002), there are few studies reported from many other countries which helped in developing a global perspective to sequestration by urban forests and carbon storage. Some of such studies are reviewed in this section.

3.1 India

The efficiency in carbon storage varies with species. A thorough study was conducted in documenting the carbon sequestration potential of various Indian vegetative species (Negi et al., 2003). Normalized difference vegetative index and carbon storage potential of urban forest in Kochi city, India, are shown in Fig. 2. The city of Kochi, India, is situated at 9° 58' N 76° 13' E and has a tropical monsoon climate. Due to its seaside location and near to the equator, Kochi experiences little seasonal temperature change and moderate-to-high levels of humidity. The annual temperature ranges from 23 to 31 °C. Conifers have the highest carbon storage, followed by deciduous, evergreen, and bamboo trees. Many studies from India (Negi et al., 2003; Pandya et al., 2013; Ugle et al., 2010) show that native Indian

trees such as *Azadirachta indica* (Neem), *Tamarindus indica* (Tamarind), *Ficus religiosa* (Peepal) and *Madhuca longifolia* (Mahua) are more efficient in carbon sequestration. It has also been reported that native and evergreen species showed better performance in biomass and carbon sequestration over the introduced and deciduous species (Majumdar & Selvan, 2018).

A study conducted at Vadodara city (Kiran & Kinnary, 2011) found that the trees planted on the road sides remove 73.59 tonnes of CO₂ representing 22% estimate of total CO₂ production of a city which is around 159.47 tonnes. Street vegetation in this populated city of Gujarat turns out to be an effective way of offsetting urban carbon emissions.

While inventorying all trees having size larger than 5 cm diameter at breast height (dbh) from a one-hectare area of the Cooum river bank at Chennai city (Pachiappan & Ushalaya, 2013) found that about 2 tonnes of carbon was sequestered by the urban forest in a year. The study indicates that existence of a greater number of small trees with lower mean stem growth contributes less to the biomass as well as carbon sequestration, compared to bigger and high growth rate trees. Another study (Niranjana & Viswanath, 2005) found that a 20-year-old silver oak shade tree may sequester up to 41.8 Mg/ha of carbon, indicating that the carbon stock in young urban trees is insignificant. Carbon sequestration, on the other hand, is dependent on tree growth rates because it reflects a possible increase in biomass.

Pune city in India has been reported to have very less area of urban forest available for sequestration in comparison with the amount of carbon emission from the city. The standing biomass of the above-ground woody sections of the trees in Pune City was determined by Waran & Patwardhan, 2001. The city of Pune and its many sectors have a carbon emission and sequestration potential of 15,000 tonnes per year. The city of Pune releases 780,000 tonnes of carbon each year, but its trees only absorb 2% of that, resulting in a 98-percent atmospheric overload (Waran & Patwardhan, 2001).

3.2 China

In 2005, the Beijing city had 2.4 million trees with most of them having smaller diameter (Yang et al., 2005). Twenty-nine percentage of trees, which was dominated by a few species, were in poor conditions and carbon dioxide (CO₂) stored as biomass amounted to about 0.2 million tonnes. According to a recent study (Tang et al., 2016), the carbon sequestration of Beijing's urban street trees is roughly 33% to that of China's non-urban woods. However, the amount of CO₂ sequestered by street trees in Beijing's entire urban district was only 0.2% CO₂ emissions.

China's industrial cities, such as Hangzhou and Shenyang, have analysed the potential of urban trees to sequester carbon (Liu & Li, 2012; Zhao et al., 2010). Hangzhou's urban woods were found to yearly neutralize 18.57 percent of the carbon generated by industrial companies and store 1.75 times the amount of carbon emitted by industrial energy consumption within the city through sequestration. Carbon sequestration neutralized 0.26 percent of yearly carbon emissions in Shenyang, with urban woods storing 3.02 percent of annual carbon emissions from fossil fuel consumption.

The monetary benefits of managing urban forests, particularly street trees, were evaluated in a field sample-based study conducted in Dalian's main urban region (Wang et al., 2018). The i-Tree programme (Nowak et al., 2013) was used to inventory 3178 trees, which employed information such as species, diameter at breast height (dbh), crown breadth, condition, and other associated parameters as input. The annual functional benefits of Dalian's street trees were estimated to be worth US\$4.9 million, with a \$4.5 million benefit per tree

for carbon storage alone. When examining the monetary benefits connected with energy savings, carbon reduction, property value, and storm water run-off reduction, the study concludes that the benefits created by urban trees are worth their management expenses. Air pollution reduction is also an important aspect of this study because it is linked to health risks and death (Gupta et al., 2021; Nair et al., 2021).

Carbon stock in urban forests includes carbon in soil as well as in tree biomass. The carbon stock in soil, though limitedly studied, has a significant contribution to the total carbon stored in urban forests. A study conducted at Harbin, most populated city of China in north-eastern region, showed that soil organic carbon accumulation during the journey of urbanization accounted for over one-quarter of the stored carbon in trees (Lv et al., 2016). Urbanization encourages carbon sequestration in soil, according to the study, which examined numerous tree and soil properties by sampling 219 plots in the Harbin city's urban forest.

Over the decades, China's urban forest form and composition have changed due to rapid urbanization. Between 1972 and 2006, a research carried out in Xiamen, a city in southern China, assessed the effects of urbanization on vegetation carbon storage (Ren et al., 2011). The data were gathered from 39,723 sample plots throughout the city, which were divided into three zones: urban core, suburban and exurban. During the 34-year period, the city's carbon storage grew by 865,589.71 tonne, with 95% of the carbon storage occurring in suburban and exurban zones due to their huge areas and the adoption of reforestation projects in these two zones. As the urban core extends towards suburban and exurban zones in the future, the study explored the risk of reduced vegetative carbon storage.

All the above-mentioned studies provide insights on how best the urban forests could play a key role in carbon sequestration in a heavily industrialized country. Human disturbances due to rapid urbanization play a dominantly influencing role on spatial and temporal distribution of the vegetative carbon stock within the urban centres of China.

3.3 United States of America

In the USA, urban forests have been widely investigated and documented for their carbon storage and sequestration (McPherson, 1998; Nowak et al., 2002; McPherson, 1994; Nowak et al., 2013). Urban forests in USA cover 35% of the urban area (Nowak & Greenfield, 2013). Updated study from US cities on the urban forests showed that total tree carbon storage of 643 million tones is available with an annual sequestration potential of 25.6 million tones. According to a study conducted from 1990 to 2005 on carbon sequestration rates in US forests (Woodbury et al., 2007), carbon sequestration in the forest sector offset 10% of US CO₂ emissions in 2005.

US Forest Service Researchers have developed many useful tools such as i-Tree (i-Tree, 2010) and Urban Crowns (Winn et al., 2010) that are useful in managing and evaluating benefits of urban trees. Furthermore, programs such as the CUFR Tree Carbon Calculator (CTCC) can be used freely to estimate carbon dioxide sequestration and building cooling/heating energy effects (Aguaron & McPherson, 2012). CTCC uses data on tree size and growth collected from around 650–1000 street trees in 16 reference cities across the USA, representing each of the four climate zones. It also includes a database of top 20–25 tree species from these cities. CTCC provides information on the carbon stock and sequestration during the past year when provided with the data on tree species, climate zone, tree size (as dbh- diameter at breast height) and age of the tree. With additional inputs, CTCC

can also calculate the energy savings, in case the tree is used for shade and it also estimates the equivalent CO₂ reduced due to the energy savings.

Type of species, density, and development of urban forests could play a surprising role in carbon storage and sequestration levels. A scenario of mapping sequestration, carbon storage, and prevented emissions via energy conservation was shown in two US cities (McPherson et al., 2013). Though Los Angeles had higher tree density and size greater than Sacramento, lower wood density tree species in Los Angeles compared to high density native oaks in Sacramento made the carbon storage levels in Los Angeles to be about 53% of Sacramento's. However, in Los Angeles, due to dense development of urban forest, 14% of the trees shaded more buildings than only 2 percent in Sacramento causing higher avoided emissions from Los Angeles compared to Sacramento.

Experience from USA on terrestrial carbon sequestration by urban forests shows that expanding urban forests by converting bare land areas may not substantially enhance CO₂ sequestration than what could be achieved through emission reduction policies. However, in addition to carbon storage and sequestration, the potential benefits of urban forests for sustainability and ecosystem services should justify urban forest restoration efforts.

3.4 Europe

In comparison with the USA, there are fewer researches on the carbon sequestration capacity of urban forests in European countries. Although research (Nabuurs & Schelhaas, 2002) evaluated the carbon sequestration profiles of 16 common forest types, experimental experiments to determine the carbon sequestration rates of European trees are uncommon. As a result, the majority of reported research have approximated carbon storage and sequestration only primarily on tree cover, utilising allometric relationships defined for US trees.

Some of the studies have used the urban tree models originally developed for North America. The UFORE (Urban Forest Effects Model) and the CUFR Tree Carbon Calculator (CTCC) were utilized in a carbon sequestration study conducted in Bolzano, Italy (Russo et al., 2014). US-based i-Tree Eco model was used for carbon sequestration studies in UK (Rogers et al., 2015). Canopy cover of urban forest in Leipzig city, Germany, has been used along with land cover to estimate above-ground carbon stock (Strohbach & Haase, 2012). Urban forest with a canopy cover of 19% of the city area was found to store 11 t C ha⁻¹ in the above-ground biomass with highest values found at intermediate urbanization levels. The carbon storage as estimated from other European cities is given in Table 2.

For decades, European woods have served as a potential carbon sink. However, these forests have just begun to show signs of surpassing their carbon sink capacity (Nabuurs et al., 2013). Higher CO₂ emissions are being caused by a slowdown in tree volume

Table 2 Carbon stored in urban forests of European cities as reported in the literature

Urban area	Carbon storage/sequestration	References
Merseyside, UK	1 to 17 t C ha ⁻¹	Whitford et al. (2001)
Edinburgh, Leicester, Glasgow, Oxford and Sheffield-UK	0.01 and 0.09 t C ha ⁻¹ yr ⁻¹	Tratalos et al. (2007)
Leicester, UK	31.6 t C ha ⁻¹	Davies et al. (2011)
Torbay, UK	15.4 t C ha ⁻¹	Rogers et al. (2015)

increment, a slowdown in net forest area expansion, increased natural disturbances such as fires, and deforestation induced by urban sprawl. Regional patterns suggest that sink saturation has occurred in the West-Atlantic and Alpine regions, while sink has increased marginally in North, South, and Eastern Europe. Afforestation is commonly thought of as a way to mitigate the effects of deforestation. It should be noted, however, that destruction of one hectare of established forest emits roughly 65 tonnes of CO₂, whereas a hectare of new forest can only store 1–2 tonnes of CO₂ each year. Review of carbon mitigation potential of different forest ecosystems including urban forests in Italy (Calfapietra et al., 2015) identified trees outside forests as a sensible carbon stock, especially in the context of urban environments.

3.5 Development of allometric relationships

Allometric relationships are key in estimating biomass quantity and carbon sequestration potential. These relationships estimate biomass volume from tree height and stem diameter measurements (Chave et al., 2005). Accuracy of these relationships mostly depends on the region for which they have been developed and also the tree characteristics that have been studied. Many studies across the globe have reported developing allometric relationships in connection with estimation of carbon storage with trees. These studies have developed models that use only diameter (*D*) as the sole parameter (Brown, 1997) or that use both *D* and height (*H*) (Basuki et al., 2009; Chan et al., 2013) or those which use *D* and *H* along with wood density (WD) (Chave et al., 2014; Djomo et al., 2010), while others have compared various allometric relationships and found that including WD improves the accuracy in determining above-ground dry biomass quantity (Pothong et al., 2021). It was reported that *D* and *H* can predict above-ground biomass more precisely for smaller trees in deciduous forests of India and also that root-to-shoot ratio also helps in estimating below ground biomass (Pati et al., 2022). However, allometric relations that are derived for forest grown trees have limitations to be applied on urban trees as they are more open grown.

3.6 Carbon sequestration in different regions of India

Due to the diverse climatic conditions in the Indian subcontinent, urban green areas vary in different regions of the country. The location of the region dictates the climatic conditions of that region. For an instance, the coastal areas like Konkan Ghats fall under hot and humid conditions, while the climate in Deccan plateau varies from semi-arid to tropical condition.

This study covers five regions of India, namely Navi Mumbai, Nagpur, Bengaluru, Leh and Delhi. All these regions are not only in different parts of the country but also in different climatic zones. Though Navi Mumbai and Nagpur fall in the same state, the climatic of Navi Mumbai is maritime tropical throughout the year, while Nagpur has a tropical savannah climate with prominently dry and hot weather. Due to the Himalayas climate in North India is mostly on the colder side. Hence, the climatic conditions in Delhi vary from monsoon-influenced human subtropical and semi-arid. However, Leh being a cold desert, the climate here is mostly cold and dry. Moving to southern part of the country, Bengaluru city falls at the tip of Western Ghats; hence, the climate here is tropical.

Three of the five cities are metropolitan cities and are highly populated and technologically advanced. Many industries have also been set up in these cities, which consequently increase the carbon emissions (Bherwani et al., 2020a, 2022). Urban forests play a huge

role in mitigation of these emissions. Hence, this study aims to evaluate and compare the carbon sequestration due to urban green spaces in these five cities and in addition defines the relation between climatic conditions and urban green spaces.

4 Methodology

In recent times, carbon dioxide emissions and climate change are two of the most critical problems. Carbon sequestration is the process of storing CO₂ or any other carbonaceous compound in order to reduce greenhouse gases, as it is framed in one of the key clauses of the Kyoto protocol. CO₂ is absorbed from the environment through ecological, biochemical, and physical processes. Carbon is a constituent of all biological living forms, and trees use it to develop limbs, bottoms, stems, branching, and leaves. Because they employ the CO₂ released for photosynthesis, urban green spaces play a critical role in carbon sequestration. By sequestering carbon and lowering pollution, urban trees play a vital ecological role in cities (Bherwani et al., 2021b; James et al., 2018). The current research focuses on estimating the urban green spaces and the amount of CO₂ sequestration in India's urban forests. The assessment is done by using NDVI. The monetary valuation of the ecosystem services is also assessed for the green spaces. The procedure of various aspects used is further detailed below:

4.1 Normalized difference vegetation index (NDVI)

Vegetation indices are utilized in solutions that show changes in land cover and forest cover, as well as for global vegetation monitoring. Vegetation influences the energy balance at the earth's surface and inside the atmospheric boundary layer, which often mitigates local temperature extremes. Vegetation cover is vitally important to animals because it releases oxygen and sequesters carbon, protects soil from erosion, turns solar energy into biomass, and provides the base of all food chains (Cavicchioli et al., 2019). The normalized difference vegetation index (NDVI) maps for India are generated for this study for the year 2018–2020 to better comprehend the vegetation. Studies have found that the NDVI is effective for expressing vegetation status and quantified vegetation attributes because of its widespread use and popularity over other methods such as enhanced vegetation index (EVI), green normalized difference vegetation index (GNDVI), chlorophyll vegetation index (CVI) (Muhsoni et al., 2018). Studies have shown that NDVI is useful for identifying different vegetation types, such as the chlorophyll concentration in leaves, plant productivity, LAI (leaf area index), biomass, fractional vegetation cover, and plant stress. It is also useful for differentiating between evergreen and seasonal forest types, savannah, dense forest, non-forest, and agricultural fields (Huang et al., 2021)

The Landsat data for this investigation comes from the United States Geological Survey's (USGS) Earth Explorer (Ganie & Nusrath, 2016). NDVI data are provided by satellite images such as Landsat 7 and Landsat 8 and data are downloaded from USGS Earth Explorer. After downloading the satellite image raw data, atmospheric correction and pre-processing are done through ARCGIS software. ArcGIS 10.6.1 platform is used to calculate the normalized difference vegetation index.

Landsat data are provided with a spatial resolution of 30 m, which is suitable to prepare NDVI maps for cities having small area with better clarity. Monthly data are downloaded to prepare the average map for the year. By utilizing the mosaic to the new raster tool in

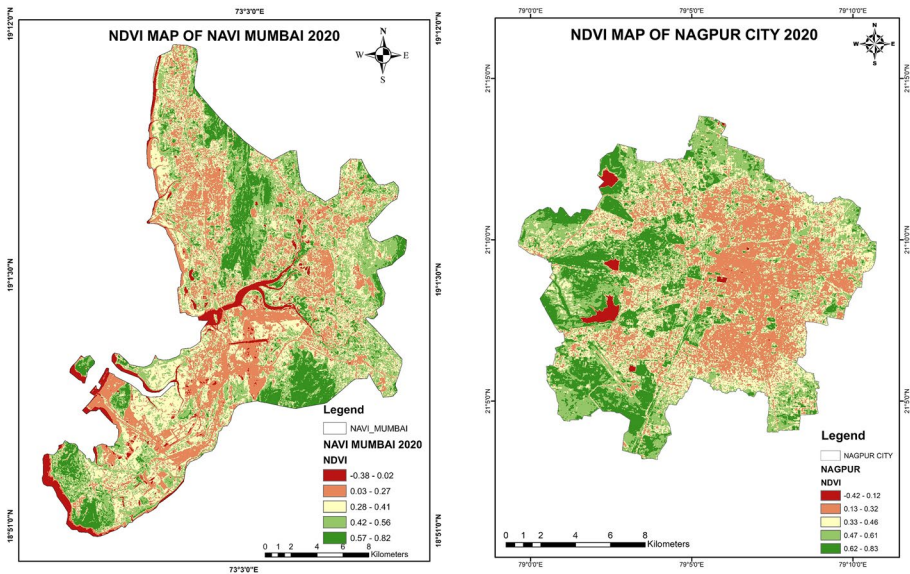


Fig. 3 NDVI Maps of Navi Mumbai and Nagpur for 2020 data

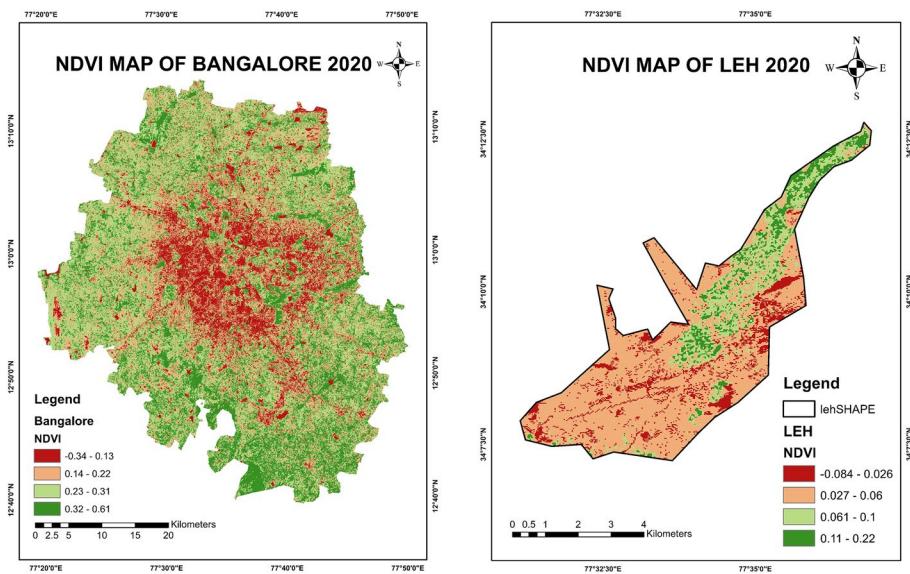


Fig. 4 NDVI Maps of Bengaluru and Leh for 2020 data

Arc-GIS, all tiles were blended into one. The values were reclassified into three categories as no vegetation cover, less vegetation, and dense vegetation and the NDVI map of Navi Mumbai, Nagpur (Fig. 3), Bengaluru and Leh (Fig. 4) and Delhi (Fig. 5) was prepared for India for the year 2020.

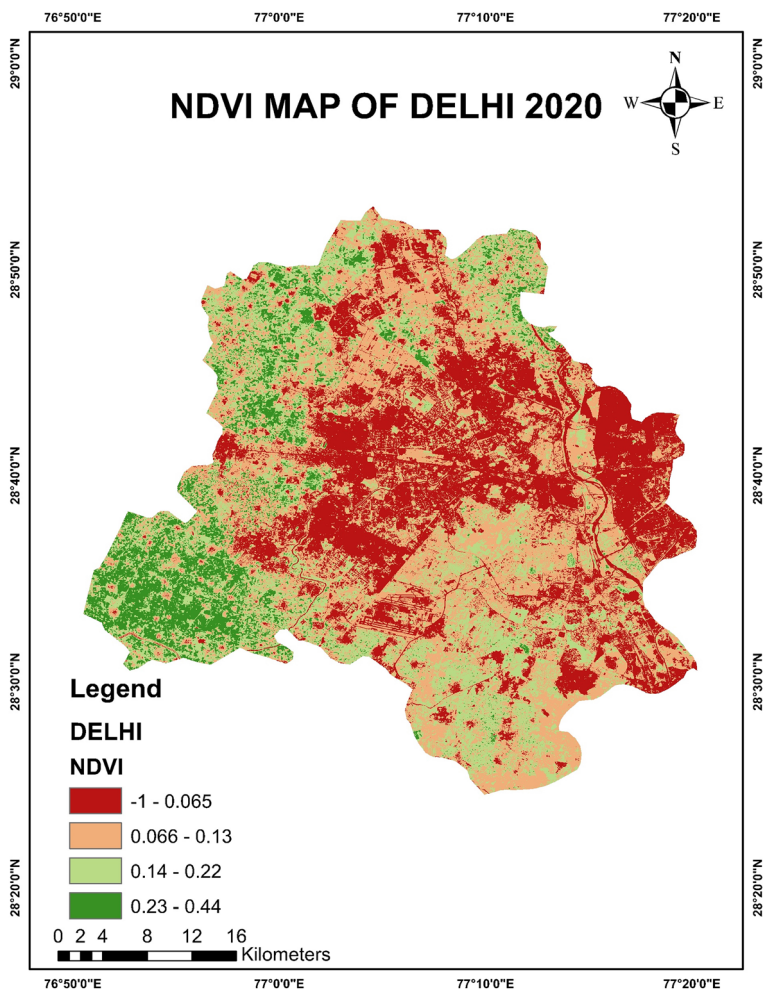


Fig. 5 NDVI Map of Delhi for 2020 data

Remote sensing techniques have a lot of promise for long-term monitoring of urban forest areas and carbon stock assessment (Huete, 2012; Myeong et al., 2006; Rao et al., 2013). Vegetation indices obtained from satellite remote sensing data have long been regarded as a main source of information on the biophysical features of vegetation across broad areas (Shojanoori & Shafri, 2016). Although there are various ways for generating vegetation indices, in applied remote sensing, the c (NDVI), which uses two spectral bands, near-infrared (NIR) and red, is the most extensively utilized. Equation 1 is used to calculate NDVI.

$$NDVI = (NIR - Red) / (NIR + Red) \quad (\text{Singh and Chand, 2012}) \quad (1)$$

The classification of area done as low, medium, and high vegetation based on NDVI range is shown in Table 3. The NDVI values near zero and decreasing negative values

indicate nonvegetated features, such as barren surfaces (rock and soil), water, snow, ice, and clouds. (Hashim et al., 2019).

4.2 Estimation of carbon storage in urban forests

In an urban setting, how much carbon can trees sequester? The assessment of the biomass of the unit under investigation is the first step in carbon quantification in terrestrial systems (e.g. a single tree). A tree's biomass, which rises with its diameter, height, and canopy spread, is proportional to the amount of carbon it stores. The quantity of carbon stored in a city grows as tree density and canopy cover increase (McPherson, 1994). Carbon sequestration, on the other hand, is the annual rate of increase in the carbon store in trees and is influenced by the urban forest's species and age structure.

In terrestrial vegetation, the standard conversion of biomass dry weight to carbon weight is: $\text{carbon} = \text{biomass} \times 0.5$. To calculate the weight of CO_2 sequestered in the tree, multiply the weight of carbon in the tree by 3.67. **An average tree can sequester about 22 kg of CO_2 equivalents (CO_2e) in an year** (McHale et al., 2009). The rate of carbon sequestration is determined by the tree species' growth characteristics, growth conditions, location of planting, and wood density. It is at its peak between the ages of 20 and 50 for trees.

Biomass quantity of individual standing trees is usually estimated using allometric relationships (i.e. estimation of biomass volume from tree height and stem diameter measurements) developed using finite number of samples from a limited region (McHale et al., 2009). If we use allometric relations that are derived for forest grown trees, it may cause discrepancies while applying to urban trees which are more open grown. Also, these relationships may cause errors if applied beyond the region(s) for which they were developed, and beyond the range of diameters that were sampled. This makes the estimation of carbon stock from allometric relations laborious and expensive especially when needs to be applied to wider regions.

Absorption of carbon by trees aids to reduce the carbon dioxide in the atmosphere termed as carbon sequestration. Increase in ability of forest to sequester emitted CO_2 can enhance other ecosystem services such as water quality and improved soil and air. For the assessment of total carbon sequestration by urban forests, details such as area covered by urban forests and available biomass of India are assorted and the carbon sequestration which refers to the amount of fixed CO_2 absorbed by various vegetation's in an area every year is used for calculations. The resulting vegetative area from the NDVI map prepared by

Table 3 Classification based on NDVI range

Classification	Classes	Range of NDVI
No vegetation	Built up, Barren land	- 1 to 0.1
Medium vegetation	Grasslands and Shrubs	0.2 to 0.5
Dense vegetation	Tropical and Temperate Forest	> 0.5

using remote sensing data is used for calculating total biomass in the area using Eq. 2 (Ren et al., 2011).

$$\text{Total biomass} = \text{SBM} \times A \quad (2)$$

The standing biomass above ground is taken from previous studies. Total carbon absorbed by the urban forests can be evaluated from the total biomass quantity as represented in Eq. 3. In terrestrial vegetation, the standard conversion of biomass dry weight to carbon weight is:

$$\text{Carbon absorbed (TC)} = \text{Total biomass} \times 0.5 \quad (3)$$

To calculate the weight of CO₂ sequestered in the tree, multiply the weight of carbon in the tree by 3.67

The total carbon sequestered is calculated by using Eq. 4.

$$\text{CS} = \text{Carbon absorbed} \times \frac{44}{12} \quad (4)$$

where CS = region's carbon sequestration, A = area of urban forest, SBM = standing biomass, i.e. total biomass of the given area at the present time, TC = amount of carbon absorbed by vegetation in one year.

4.3 Estimation of urban green spaces

Urban green spaces are providing immense ecosystem services to the urban community. In this study, urban green spaces play a key role in the carbon sequestration in order to reduce carbon emissions and mitigate severe environmental effects (Mader, 2020). For this study market price method, value transfer method and social cost of carbon are used for estimation. For the purpose of deriving the values of carbon sequestration, a wide range of research studies were reviewed. The monetary values of this service are obtained from several studies, and a wide range of carbon sequestration by urban green spaces is evaluated. These values are then applied to the study area through an approach known as "Value Transfer Method" (VTM). The VTM was developed to gauge the economic worth of ecosystem services. This strategy is based on the practise of transferring data from previously completed studies to new locations or contexts. The values thus selected are inflated, and suitable Purchase Power Parity (PPP) is used in order to estimate the values for India for the year 2022. PPP is the rate at which one country's currency would need to be exchanged for another country's currency in order to purchase the same quantity of goods and services in both (IMF). For market price method, the market price of carbon stored is considered from the literature and then similar process of VTM was used for the study. Social cost of carbon (SCC) is used to calculate the monetary cost of environmental damage due to CO₂ emissions. It highlights the economic consequences of rising CO₂ emissions as a result of climate change. SCC uses the economic cost of carbon and carbon emitted to evaluate the total economic impact generated due to CO₂ emissions. The SCC methodology addresses the social and environmental damages occurring due to carbon emissions. They make a comprehensive assessment of damage costs due to climate change and many other factors like variation in net agricultural productivity, human well-being, monetary losses due to flood risks and changes in energy system costs, which include rise in cost of air-conditioning and reduction in cost of heating (Bherwani et al., 2019, 2020b). The current study also makes use of the same approach for assessment. For this purpose, the quantum of CO₂ emitted from every source in India is considered for SCC calculation.

The impact of these emission reductions due to urban green space is monetarily quantified using the SCC equation 5 (Bherwani et al., 2019) provided below:

$$\text{SCC} = \text{Quantity of CO}_2 \text{ emitted (tCO}_2\text{)} * \text{Cost of CO}_2 \text{ per tonnes} * \text{Purchasing power parity (PPP)} \quad (5)$$

5 Analysis and results

The analysis of the different parameters done with the estimation methods that are discussed above in the methodologies is detailed below in subsequent section.

5.1 Normalized difference vegetation index (NDVI) of cities

The map of NDVI for cities including Navi Mumbai, Nagpur, Bengaluru, Leh, and Delhi for the year 2020 is shown in Figs. 3, 4 and 5.

The analysis of Table 4 shows vegetation cover area for the cities including Navi Mumbai, Nagpur, Bengaluru, Leh and Delhi for the year 2020.

5.2 Quantification of carbon absorbed in urban forests

The quantification of carbon stored is calculated using the vegetative area found through remote sensing and standing biomass is considered from literature and the carbon sequestration is calculated as shown in Table 5.

5.3 Valuation of urban green spaces

The valuation is done by using market price method, value transfer method and social cost of carbon. For market price method and value transfer method, the values are considered from the literature and by using VTM; the previous estimates are transferred to the current study. The values thus selected are inflated and suitable Purchase Power Parity (PPP) are used in order to estimate the values for India for the year 2022.

5.4 Social cost of carbon (SCC) method

The values for cost of CO₂ per tonnes emitted to the atmosphere are taken as per USEPA for the year 2020. Monetary assessment is carried out for an average discount rate of 3 % scenario. The values used for the evaluation have been rounded to the nearest whole digit and are shown in Table 6.

The emission of CO₂ for the study area (Bengaluru) is taken from aforementioned calculation, and the SCC for an average impact scenario of 3% yielded a value of 6537.15 million dollars for the year 2022. This monetary quantification of reduced

Table 4 Details of NDVI map for the year 2020

Sr No.	CITIES	NDVI values	Area (Sq. Km)			
			Barren	Low vegetation	Medium vegetation	High vegetation
1	Navi Mumbai	− 0.38 to 0.82	106.46	133.26	117.38	88.61
2	Nagpur	− 0.42 to 0.83	1.914	0.54	96.55	121.06
3	Bengaluru	− 0.34 to 0.61	25.32	154.57	202.47	1813.44
4	Leh	− 0.084 to 0.22	4.03	17.06	4.02	2.06
5	Delhi	− 1 to 0.44	164.52	477.57	413.77	442.48

Table 5 Carbon dioxide sequestration for the year 2020

Vegetation in city	Area (Sq. Km) 2020	Area (Ha) 2020	Standing biomass (t/ha)	Total biomass (tonnes)	Carbon absorbed (total biomass * 0.5)	CO ₂ sequestration (carbon absorbed * 3.67) (million tonnes)
Navi Mumbai	339.25	33,925	356.11	12,081,031.75	6,040,515.88	22.17
Nagpur	218.15	21,815	356.11	7,768,539.65	3,884,269.83	14.26
Bengaluru	2170.48	217,048	356.11	77,292,963.28	38,646,481.64	141.83
Leh	23.14	2314	356.11	824,038.54	412,019.27	1.51
Delhi	1333.82	133,382	356.11	47,498,664.02	23,749,332.01	87.16

Table 6 Estimated social cost of carbon (SCC)

Specifications	Values
Quantity of CO ₂ emitted (2020) [tonnes]	141,832,588
SCC for carbon for 2020 as per USEPA on 3% average discount rates (\$) (Bherwani et al., 2019)	42
Estimated Social Cost of Carbon—2020 (SCC) for 3% average discount rates [million \$/year]	5957
Estimated Social Cost of Carbon—2022 (SCC) for 3% average discount rates [million \$/year]	6537

CO₂ emission reveals the avoided damages to the environment and change in climatic conditions, which would occur due to these reduced CO₂ emissions, have huge worth. According to the findings, the rise in carbon dioxide emissions can be minimized sequestration of this released CO₂ by increasing the urban green spaces. As a result, in the current situation, there is a need to increase urban green areas.

The similar procedure was followed for valuation of the absorbed carbon dioxide by MPM, SCC and VTM approaches. Table 7 shows the valuation of absorbed carbon dioxide using the MPM, SCC and VTM methods, with the values rounded to the nearest whole digit. In the same line with the previous results, the highest valuation was obtained for Value Transfer Method followed by Social cost of Carbon and Market Price Method. Leh is a small region in terms of area and in addition being a dry region has relatively lower green cover. Hence, carbon sequestration is minimum in case of Leh city. Highest carbon sequestration and valuation values were observed in Bengaluru followed by Delhi.

Similar to the previous calculations, valuation was carried out for three more countries: USA, China, and Europe. The carbon sequestration data provided in the case studies were used for calculating social cost of carbon for the year 2022 (Table 8).

Table 7 Valuation by different approaches for the year 2022

Valuation method	Navi Mumbai	Nagpur	Delhi	Leh	Bengaluru
Market price method (million \$)	807	519	3173	55	5164
Social cost of carbon (million \$)	1022	657	4017	70	6537
Value transfer method (Million INR)	23,092	14,849	90,788	1575	147,737

Table 8 Valuation for different cities all over the world

Country	Carbon sequestrated (tonnes)	Valuation by SCC, 2022 (million \$/year)
USA	25,600,000	1180
Xiameng, China	25,458.52	1.2
Leipzig, Germany (Europe) (Strohbach & Haase, 2012)	22,891	1.1

6 Conclusion

Urban forests have direct and indirect contribution to carbon sequestration. Direct is by absorbing CO₂ for photosynthesis of biomass. Indirect contribution is due to the cooling effect they offer due to the shade, thereby reducing energy use and associated carbon emissions. Nurturing urban forests and reducing the chances for its degradation therefore complement if not enhance the carbon reduction efforts of cities. Carbon absorbed by the urban forests predominantly depends on the area of the region of study.

Under this study, potential of five cities, namely Navi Mumbai, Nagpur, Bengaluru, Leh, and Delhi, was investigated in terms of carbon sequestration. It was observed that Bengaluru showed highest carbon sequestration (141.83 MT) and carbon sequestration in the city of Leh was comparatively very low (1.51 MT). Leh is said to be a cold desert; hence, the urban forest cover in the region is very low, which consequently leads to lower carbon sequestration. Nagpur and Navi Mumbai are both in Maharashtra, but they have quite diverse climate zones, which is one of the most significant elements to consider when calculating carbon sequestration. Carbon sequestration in Navi Mumbai is higher (22.17 MT) than that of Nagpur (14.26 MT) because of greater area and because of wet climatic conditions which enhance the urban green spaces.

The present study also evaluated the economic importance of urban green spaces as they play a key role in the carbon sequestration in order to reduce carbon emissions and mitigate severe environmental effects. The methodology of such valuation of urban green spaces in the present study included Market Price Method (MPM), Social Cost of Carbon (SCC) and Value Transfer Method (VTM). The SCC approach used for valuation shows the highest cost of CO₂ sequestered, while the MPM provides with the lowest value for all the five cities. Lahoti et al. (2020) investigated the existing species diversity, composition, and structure in different types of greens found in Nagpur and concluded that urban green spaces are carbon stock reservoirs. A non-destructive approach is used with a species-specific equation to measure biomass and observed that biomass was abundant in roadside plantations (335 t ha⁻¹) and playgrounds (324 t ha⁻¹) and that trees with larger girth sizes were the main contributors. According to a study that examined trends in vegetation greenness patterns across six major Indian cities using remote sensing data and observed from the entropy index, Bangalore exhibited significantly less vegetation dispersion than Hyderabad and New Delhi (Prasad et al., 2003). Davies et al. (2011) investigated the quantity and geographical patterns of above-ground carbon deposited in Leicester, a British city, and vegetation throughout the entire urban region and observed that 231,521 tonnes of carbon are stored in Leicester's above-ground vegetation, which is comparable to 3.16 kg C m⁻² of urban area, with the largest above-ground carbon density linked with regions of tree cover on publicly owned/managed locations at 28.86 kg C m⁻². Nawar et al., (2022) performed NDVI analysis to calculate the total change in vegetation. After reviewing the literature studies, this study also calculates vegetation cover using the NDVI technique. For Navi Mumbai, Nagpur, Delhi, Leh and Bengaluru, the absorbed carbon dioxide by MPM method is to be around 807, 519, 3173, 55 and 5164 million US \$, respectively. Using the SCC approach, the value of the absorbed carbon dioxide is to be around 1022, 657, 4017, 70 and 6537 million US \$, whereas using the VTM method it could be around 23,092, 14,849, 90,788, 1575 and approx. 147,737 INR, respectively. The valuation is directly proportional to the amount of carbon sequestered. Therefore, as predicted, the valuation of sequestered carbon was highest for the city of Bengaluru followed by Delhi, Navi Mumbai, and Nagpur and lowest for the dry city of Leh.

Muhsoni et al. (2018) and Huang et al. (2021) discovered that the NDVI is useful for communicating vegetation status, vegetation types, such as the chlorophyll concentration in leaves, plant productivity, LAI (leaf area index), biomass, fractional vegetation cover, plant stress and quantitative vegetation properties because of its extensive use and popularity over other methods such as the Enhanced Vegetation Index (EVI), Green Normalized Difference Vegetation Index (GNDVI), and Chlorophyll Vegetation Index (CVI). Therefore, in order to better understand the vegetation, normalized difference vegetation index (NDVI) is developed for this study for the year 2018–2020 and it is observed from the NDVI maps that Bengaluru has highest vegetation cover and Leh has lowest vegetation cover. This monetary quantification of reduced CO₂ emission reveals that increasing urban green spaces has high economic worth in terms of their carbon storage and sequestration potential.

7 Policy recommendations and scope for future work

Managing healthy forests in most dense urban areas is an effort worth taking for the benefits it offers to the city dwellers. Degradation of urban forests can occur due to urbanization, encroachment, neglect of natural areas, fragmentation and spread of invasive species. Among these, reduction in tree cover due to infrastructure developmental activities is one of the major reasons for urban forest reduction. Reduction of urban tree cover due to major infrastructure projects can affect terrestrial carbon sequestration potential of cities. Increased population pressure is driving urban areas, especially in the developing countries, to expand and create amenities for the inhabitants. It is a global practice to conduct environmental impact assessment for large-scale infrastructure projects, which mostly assess the impact of proposed project on the local environment. However, the loss of urban forest cover has a significant and often underestimated climatic impact, as the carbon that would have been absorbed by the trees is now released into the atmosphere as emissions.

Most impact mitigation plans promote afforestation at alternate locations within a city to compensate for the trees that are cut for urbanization. In comparison with newly planted trees, the trees being chopped are usually larger and older, with more sequestration capacity. A recent study in Kochi, India, looked at the spatiotemporal changes in terrestrial carbon sequestration as a result of a metro rail project (James et al., 2018). During the study period, 14.51 ha of tree cover area within 60 m of the metro rail centre-line was reduced, resulting in the emission of 2316 t CO₂e (tonnes of carbon dioxide equivalent). This countered the reduction in emissions owing to the city's encouragement of public transportation. Therefore, policies on urban green spaces and trees will have to be formulated keeping in mind their ecological and climatic services.

Growing forests show a positive net annual carbon sequestration, and as the forest mature, the sequestration rate diminishes. Many large cities have discovered that urban forests do not provide considerable carbon absorption and that converting barren land to urban trees does not significantly reduce CO₂ emissions. However, when managing trees for climate change mitigation and carbon sequestration, long-term aims, numerous ecosystem services, costs, community benefits, and the preservation of existing forests should all be taken into account.

Studies from India show that the percentage area of urban forest required in any city shall be decided based on the total carbon emission from the urban activities. If the area is lesser, urban forest restoration programmes should be given priority. Sequestration would

be much more effective if native species are used. During urban development, preserving large trees or planting trees with higher growth rate enhances carbon sequestration.

Estimating the carbon stock with lesser uncertainties is important for accurate quantification of carbon sequestration. Carbon stock can vary with respect to tree species, wood density, age and growth rate. Below-ground biomass estimation and a lack of knowledge about the amount of soil carbon reservoirs in urban forests are two major uncertainties in carbon stock assessment. Root biomass varies greatly, ranging from 16 to 41% of the biomass of urban trees, whereas 61% of the total carbon in non-urban forest ecosystems is stored in the soil environment. Estimation of carbon sequestration potential could be done using field measurements and remote sensing analysis. In this study, an attempt is made to evaluate the sequestration of carbon in urban forests by examining NDVI maps that have been created at the city level in India and have previously been proposed in the USA and Europe. The study area's forest structure requires for a substantial amount of fieldwork, which takes time and effort and will be covered in later studies.

Most of the studies from around the world use the carbon stock estimation using the allometric relationships developed for the cities in USA as such extensive study is limited for other countries. Thus, there is a need for developing allometric relationships for specific regions in India for the accuracy of the estimations presented in this work.

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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